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Fundamentals of Programming

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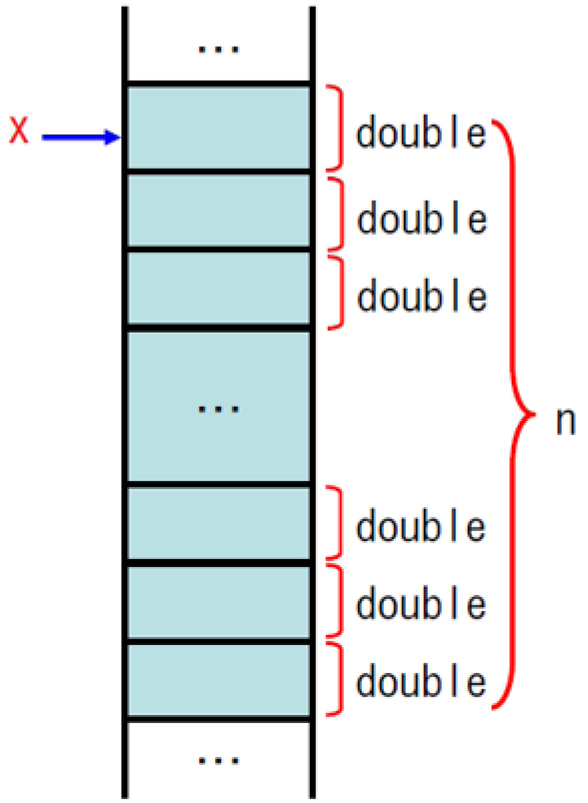
May 19th

Outline

- Malloc
- Structure

Malloc

```
double *x;
x = (double *)malloc(n*sizeof(double));
```



sizeof(double) → 8 byte
sizeof(int) → 4 byte
sizeof(char) → 1 byte

Malloc

- x

```
x = (double *)malloc(n*sizeof(double));
```

- (double)

```
int a, b;
double c, d;

a = 1.0;
b = 3.0;
c = a / b;          →  0.000000
d = (double)a / b;  →  0.333333
```

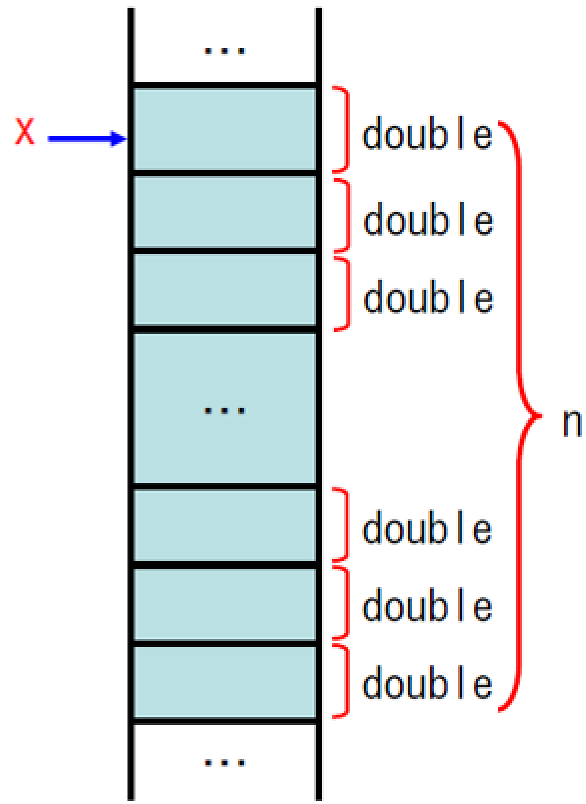
Malloc

#include <stdlib.h>

```
double *x;
x = (double *)malloc(n*sizeof(double));
```

```
x + i
*(x + i)
x[i]
```

```
scanf("%lf", &x[i]);      →
printf("%.2f¥n", x[i]);   →
```



Malloc

```
double *a, *ai;

a = (double *)malloc(n*n*sizeof(double));
if (a==NULL) {
  printf("Can't allocate memory.¥n");
  exit(1);
}
```

- Poniter ai

```
ai = a;
```

- Change line

```
ai += n;
```

Malloc

```
double *a, *ai;

a = (double *)malloc(n*n*sizeof(double));
if (a==NULL) {
  printf("Can't allocate memory.¥n");
  exit(1);
}

ai = a;
for (i=0; i<n; i++) {
  for(j=0; j<n; j++) {
    printf("a[%d][%d] = ", i, j);
    scanf("%lf", &ai[j]);
  }
  ai += n;
}
```

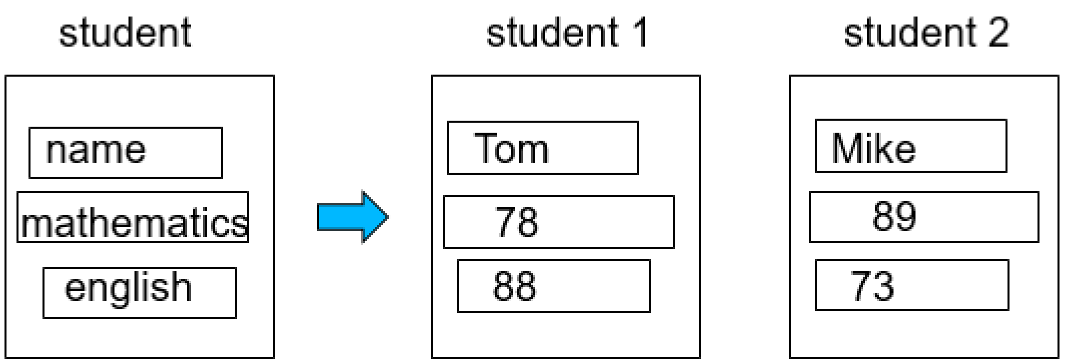
Outline

- Malloc
- Structure

Structure

- A structure is a collection of one or more variables, possibly of different types, grouped together under a single name for convenient handling

- For example, student record



Structure

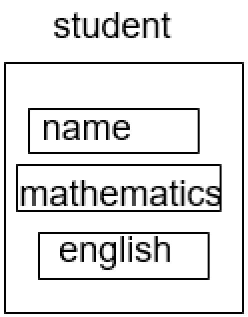
An optional name called a structure-tag may follow the word struct

```
struct structure-tag{
 member1;
 member2;
};
```

The variables named in a structure are called members

- For example, student record

```
struct student {
    char name[20];
    int eng;
    int math;
};
```



Structure

A struct declaration defines a type.

```
#include <stdio.h>
struct student {
    char name[20];
    int eng;
    int math;
};

int main(void)
{
    struct student a, b, *pa;

    ...
}
```

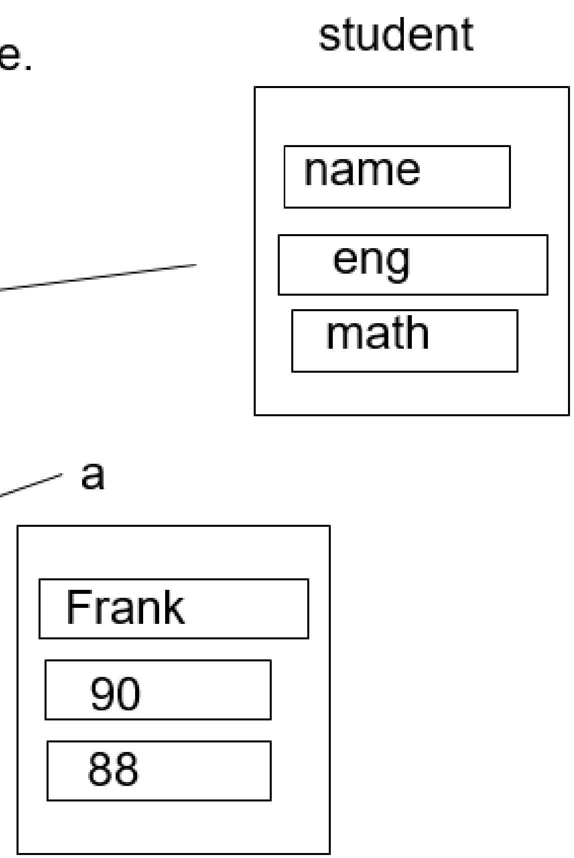
Structure

A struct declaration defines a type.

```c
#include <stdio.h>
#include <string.h>
struct student {
    char name[20];
    int eng;
    int math;
};

int main(void)
{
    struct student a, b, *pa;

    strcpy(a.name, "Frank");
    a.eng = 90;
    a.math = 88;
    ...
}
```



Structure

```
struct student a, b, *pa;
```

```
pa = &a;
```

a

pa → | name |
|---|
| eng |
| math |

```
pa = &a;
strcpy(pa->name, "Thomas");
pa->eng = 85;
pa->math = 94;
```